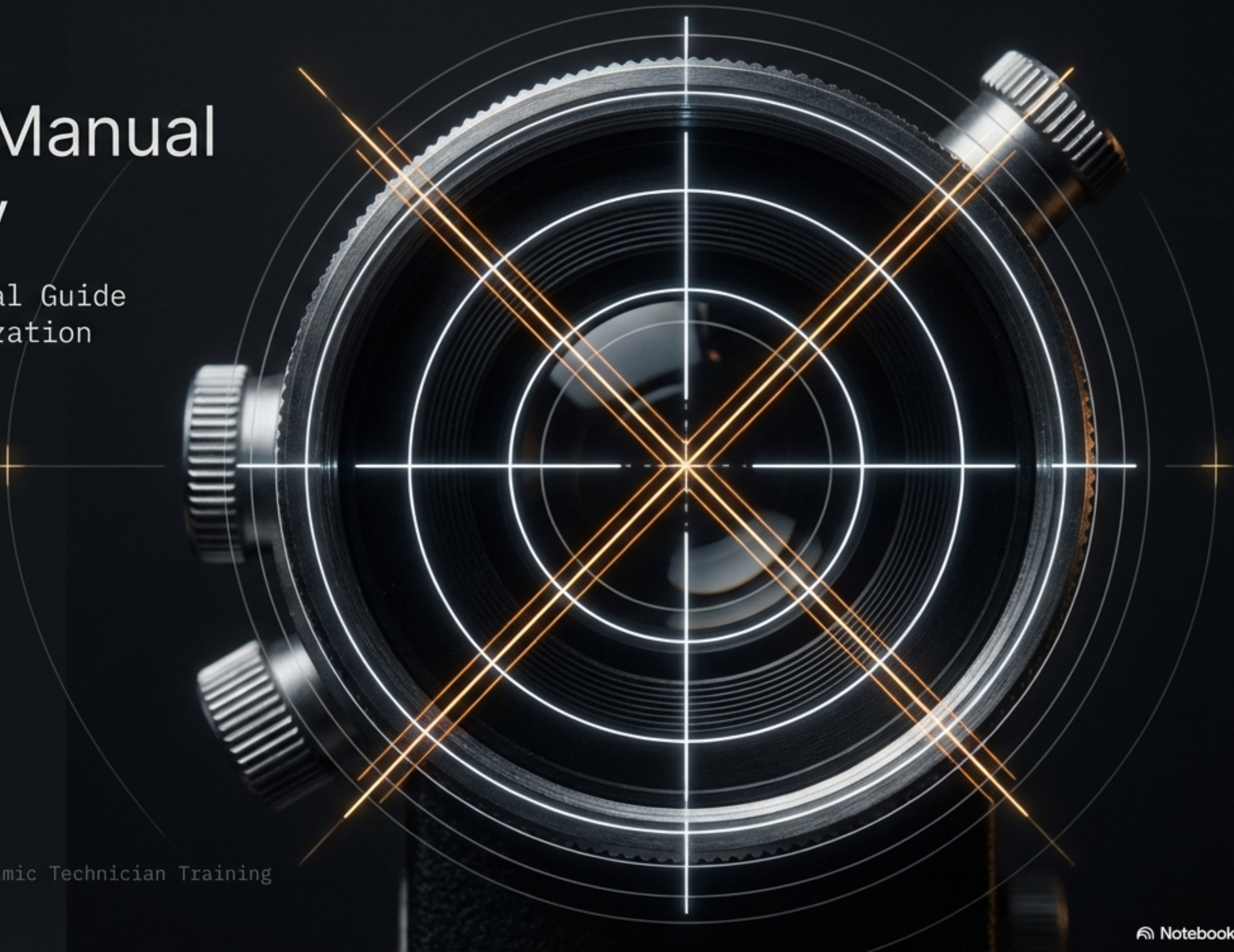


Mastery in Manual Lensometry

A Step-by-Step Clinical Guide
to Precision Neutralization



EYEPIECE:

The operator's optical interface.

AXIS ADJUSTMENT WHEEL:

Aligns the cylinder meridians.

SPECTACLE TABLE & CONTROL:

Levels the frame to ensure axis accuracy.

POWER DRUM:

The primary measurement dial (black numbers = plus, **red** numbers = minus).

GIMBAL (LENS HOLDER):

Secures the lens in the precise focal plane.

PRISM COMPENSATOR:

Decodes complex optical displacement.

Precision neutralization requires simultaneously managing the power drum, axis wheel and, and spatial alignment. It is a fully tactile, mechanical process.

Phase 1: Zeroing the Instrument



Out of Focus (Blurred)



Turn Clockwise



In Focus (Sharp)

1. Set to Zero

Turn the power drum to 0.00 and set the prism compensator to zero.

2. Blur the View

Turn the eyepiece completely counter-clockwise until the reticle stops and is entirely blurred. (Gentle motion)

3. Find the Edge

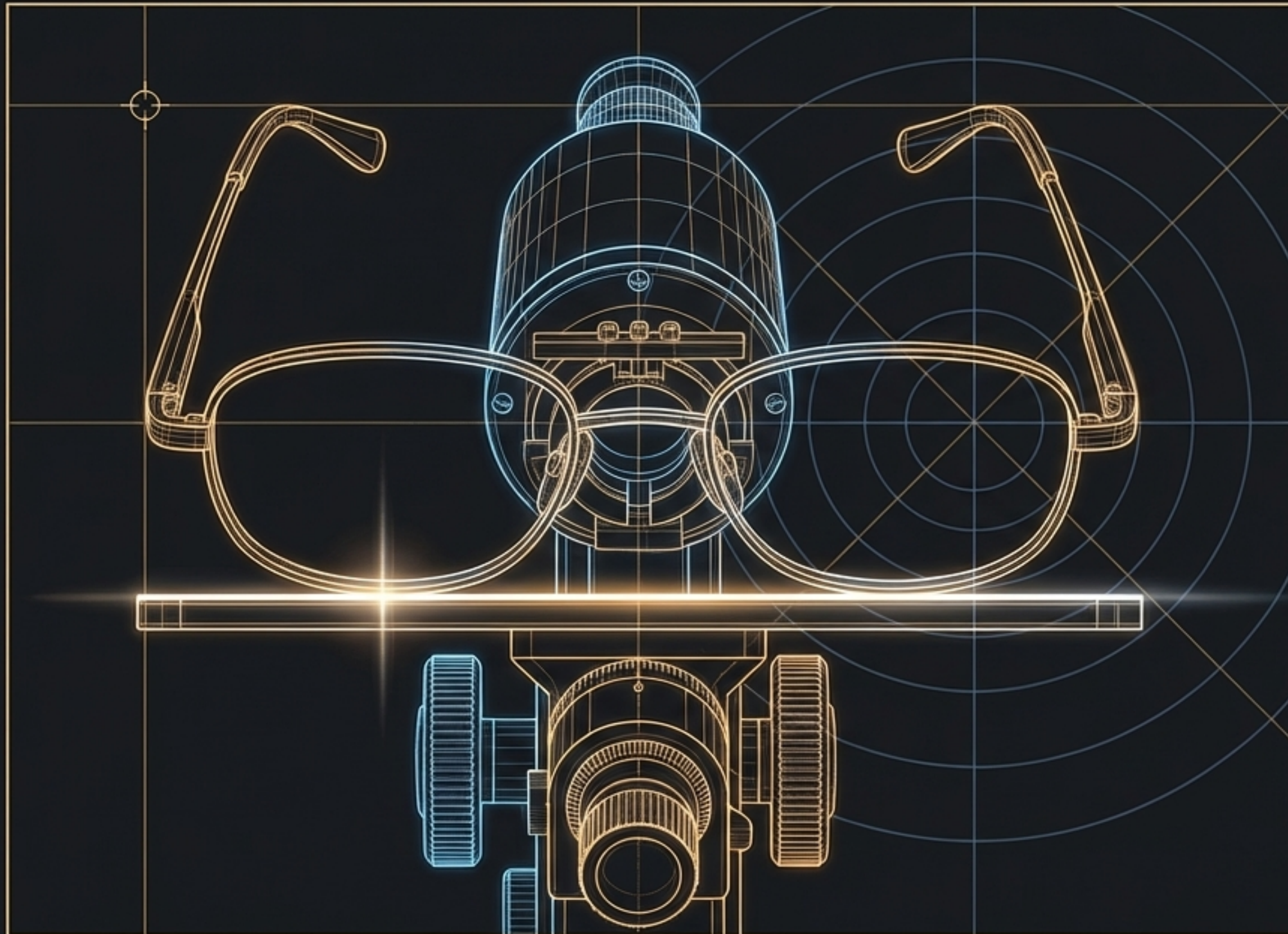
Look into the eyepiece and slowly turn it clockwise just until the reticle crosshairs are sharply in focus.



Clinical Warning Box

DO NOT OSCILLATE. Never twist the eyepiece back and forth. Oscillating forces your eye to accommodate, fundamentally corrupting every subsequent reading. Stop turning the moment the reticle is clear.

Phase 2: Positioning the Spectacles



Measure Back Focal Length:

Temples must face away from the operator. The instrument is designed to measure the back surface of the lens.

Maintain the Axis:

Both lenses must rest completely flat and flush on the spectacle table. If the frame is bent, hold the frame level manually to avoid false astigmatism readings.

Standard Protocol:

Always begin with the right lens, then proceed to the left.

Action Prompt:

Lock the lens in place gently with the Gimbal once the mires are centered in the reticle.



Decoding the Viewfinder (The Mires)

The Reticle (White):
The stationary central crosshairs, concentric rings, and outer protractor axis scale.

Sphere Lines (Amber):
Three thin, closely spaced, parallel lines.



Cylinder Lines (Neon Blue):
Three thick, widely spaced, parallel lines (sitting exactly perpendicular to the thin lines).

You are aligning two independent focal planes. The thin lines dictate the primary sphere power; the thick lines dictate the secondary cylinder power.

Step 1: Measuring the Sphere Power



1. **Start High:** Turn the power drum away from you to a high plus power (+10.00).
2. **Move to Minus:** Slowly turn the drum towards you (into the minus direction).
3. **Isolate:** Stop the exact moment the thin lines come into sharp, unbroken focus.

Reading 1: The Anchor
-1.50 Sphere

Diagnosing the Lens Type

The Spherocylinder Matrix

Condition A: The Spherical Lens



Thin AND thick lines are in perfect focus simultaneously.

Cylinder = 0.00.

The reading is complete.

Condition B: The Spherocylinder Lens



Thin lines focus, thick lines are hazy/blurred.

Astigmatism is present. You must proceed to Step 2 to find the second focal point.

Condition C: The Broken Axis



The thin lines are sharp but broken.

Correction: Rotate the outer axis dial until the thin lines are unbroken and continuous before reading the sphere power.

Step 2: Measuring Cylinder (Minus Cylinder Form)

Distance Traveled = -1.50 Cylinder



The Procedure

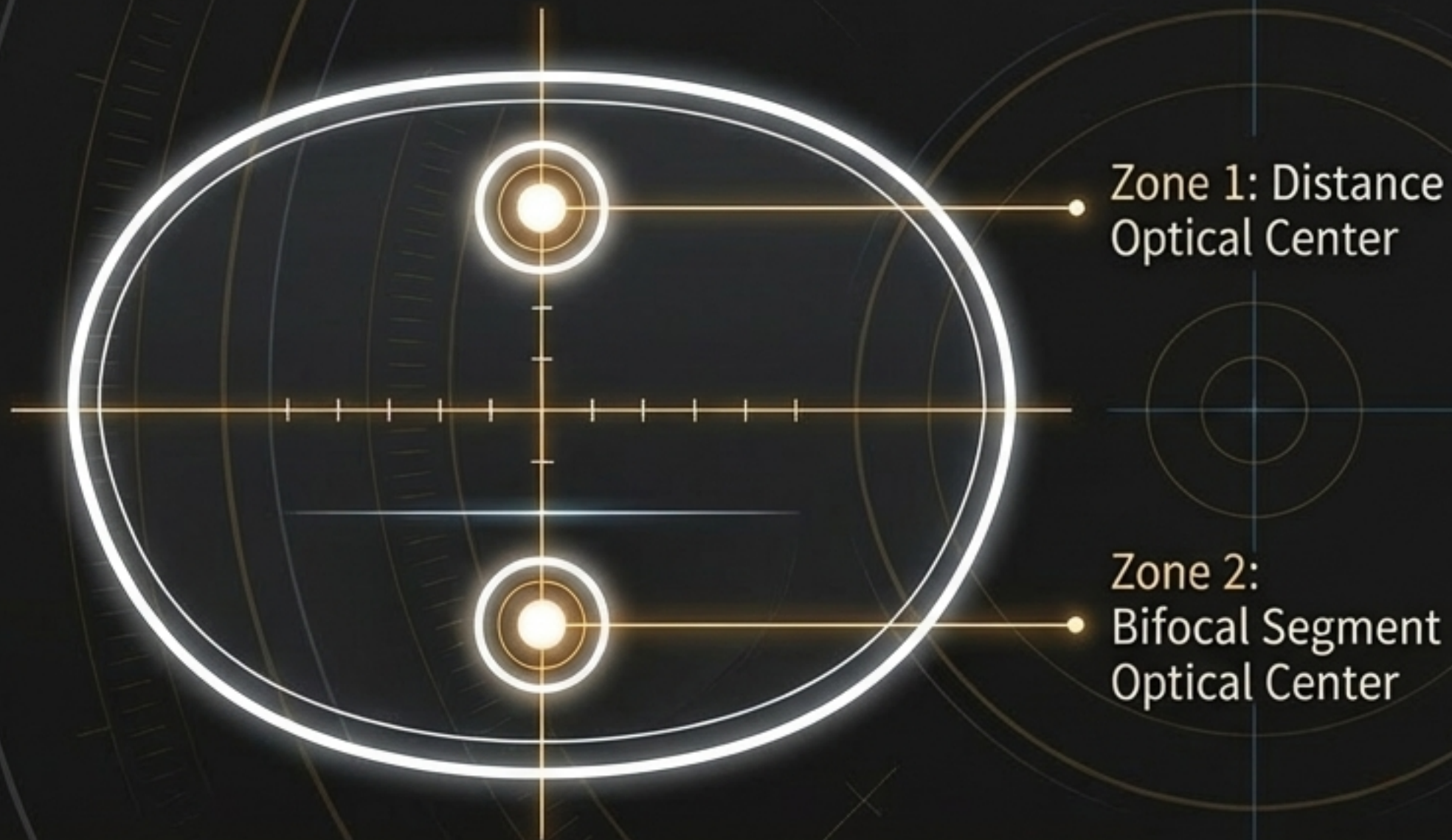
1. Leave the thin lines perfectly focused.
2. Rotate the power drum away from you (adding more minus power) until the thick lines come into perfect focus.
3. Note the axis on the axis wheel (e.g., 164°).

The Mathematical Rule & Output

The Rule: The cylinder power is the exact mathematical difference between your first reading and your second reading.

Prescription Neutralized: -1.50 - 1.50 x 164

Phase 3: Setup for the Near Add



The Procedure

Measure the distance correction normally at the primary optical center (Zone 1).

Push the glasses upward on the stage to perfectly center the bifocal segment over the lens stop (Zone 2).

Note: The mires may not align perfectly with the central crosshairs in the segment—this is acceptable. The goal is clarity, not perfect centration.

Calculating the Add Power

Distance Traveled = +2.00 ADD.



The Procedure

1. With the segment centered, rotate the power drum back towards you (into the plus direction).
2. Stop when the thin lines refocus.
3. The Add power is the algebraic difference between the original distance sphere reading and this new reading.

Clinical Tip

Tip: For trifocals, the intermediate section is typically exactly 1/2 the power of the lower reading section.

Phase 4: Understanding Prism Magnitude



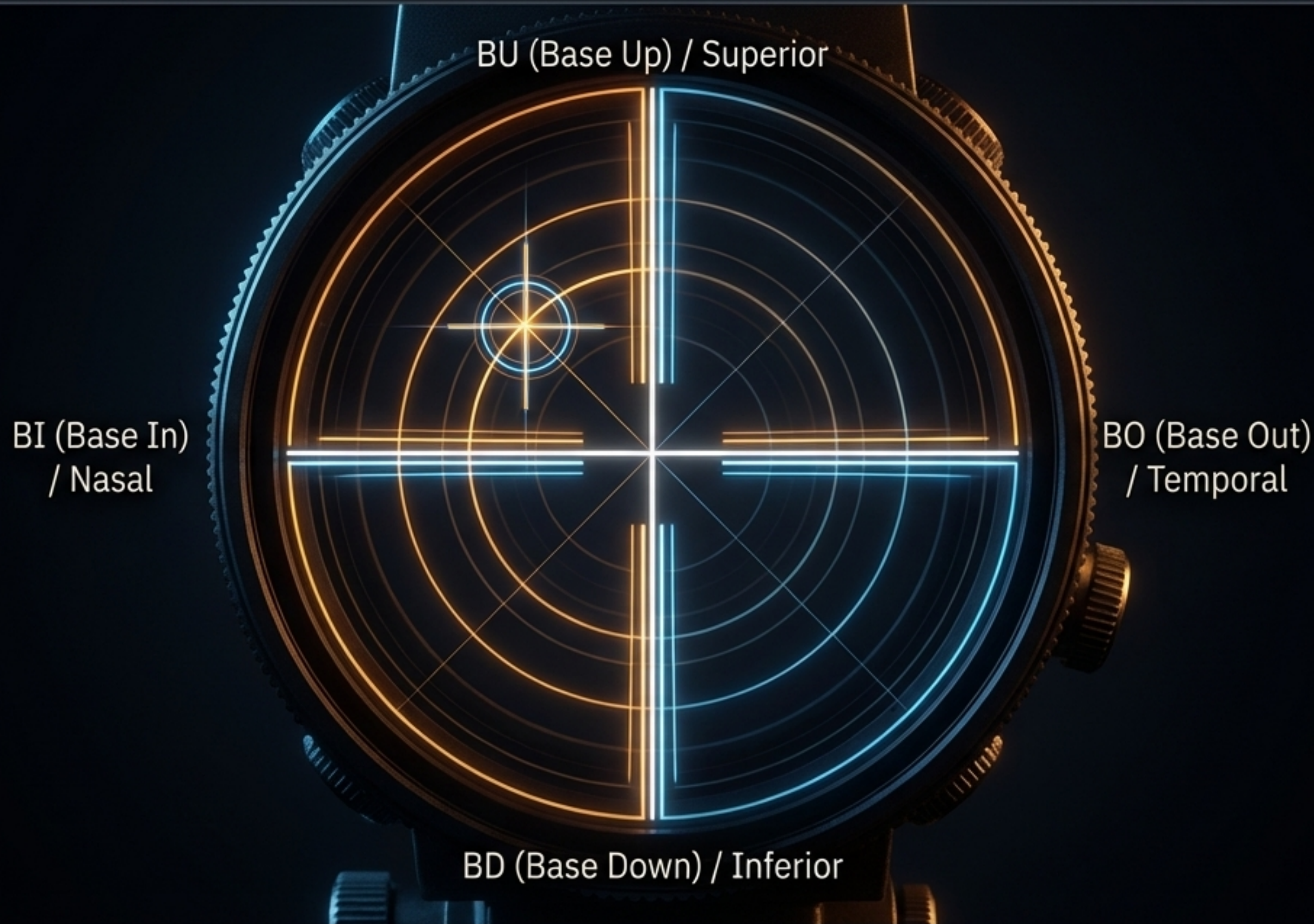
The Concept

Prism is present when the mires can be focused sharply, but cannot be centered directly on the central crosshairs.

Decoding the Rings

- Each black concentric ring represents exactly 1 Prism Diopter (1Δ).
- **Critical Detail:** The very first, unmarked ring closest to the center represents $\frac{1}{2}\Delta$.
- To find the magnitude, simply count the rings from the exact center of the reticle to the intersection point of the focused mires.

Mapping Prism Direction



Clinical Application

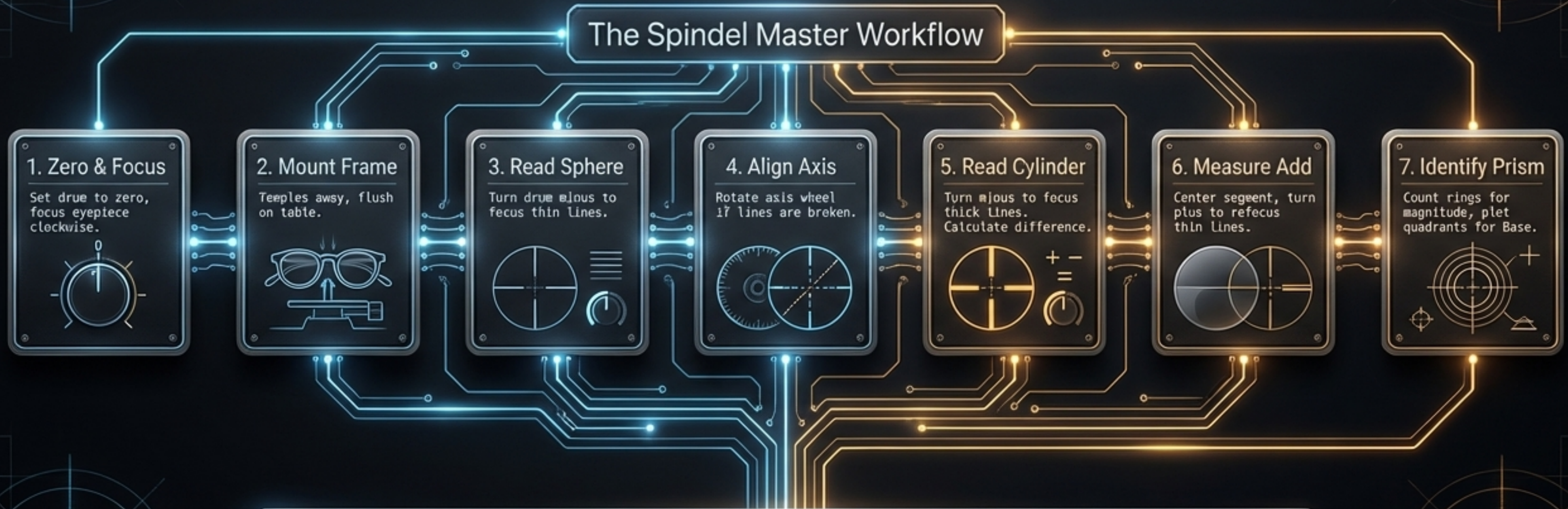
Always assess displacement relative to the patient's nose (Nasal vs. Temporal).

Example Calculation

In the visual provided, the mires are displaced 2 rings Superiorly and Nasally.

Therefore, the lens contains a compound prism of 2Δ BU and 2Δ BI.

The Pathway to Neutralization



Mastery is not just understanding the machine; it is executing the sequence with flawless, repeatable precision.